

Visual Planning: Reasoning Through Images Instead of Words

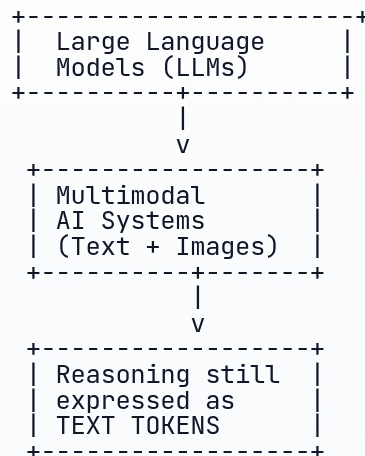
🔍 Explanation

[compass] Modern artificial intelligence systems increasingly perform reasoning tasks, but the way they structure thought is still heavily tied to language. [lens] This document explains a new paradigm where reasoning can occur visually through sequences of images rather than purely through textual descriptions. [bridge] The concept introduces visual planning as an additional reasoning channel tha





[tower] The Rise of Multimodal Reasoning Models



Problem:
Visual tasks → forced into textual reasoning



Large Language Models have dramatically expanded the ability of machines to perform reasoning tasks across domains such as coding, analysis, and planning.



Modern systems now integrate multiple data types including text and images, creating multimodal models capable of interpreting visual and linguistic signals together.



Despite these capabilities, the reasoning process itself usually remains expressed in textual form, even when the underlying task involves visual information.



This dependence on language creates a structural bottleneck because text is not always the most intuitive medium for representing spatial or geometric reasoning.

LLMs process reasoning primarily through text tokens

Multimodal models can perceive images but still reason in language

This creates limitations for spatial reasoning tasks



[map] The Limits of Language for Spatial Reasoning

SPATIAL PROBLEM

```
+-----+
|      MAZE      |
| Start → → → Goal |
+-----+
```

Natural reasoning:
VISUAL

Forced reasoning in LLM:

"move right, then up,
then left, then down..."

Result:

```
+-----+
| Spatial relations |
| become word sequences |
+-----+
```



Certain problems involve relationships that are inherently spatial, such as navigating through a maze or planning movement across a map.



When such problems are forced into text-based reasoning, the model must translate visual relationships into words before performing inference.

This translation step increases cognitive complexity for the system and may distort important spatial structures.



As a result, relying purely on language can hinder performance in tasks where



geometry, layout, and spatial intuition are central.

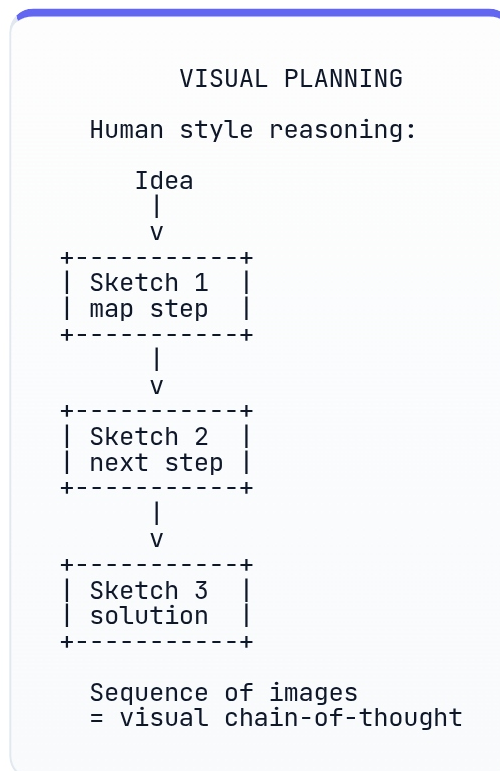
Spatial reasoning often depends on visual relationships

abc Text-based reasoning introduces translation overhead

! Important geometric information can be harder to encode in language



[canvas] The Visual Planning Paradigm



Visual Planning introduces a new way for models to perform reasoning by constructing sequences of images that represent intermediate steps of thought.



Instead of describing plans in text, the system visually sketches its reasoning process much like humans drawing diagrams or planning routes on paper.



Each image acts as a frame in a reasoning sequence, showing how the environment

evolves step by step as decisions are made.



This approach supplements traditional language reasoning by adding a visual channel where spatial logic can be represented more naturally.



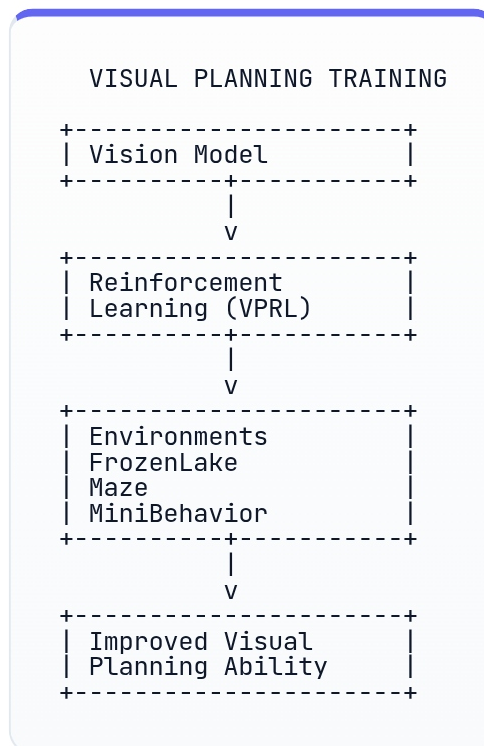
Reasoning steps are encoded as visual frames

Planning occurs through sequential image transformations

The method mirrors human sketch-based reasoning



[engine] Training Visual Planning with Reinforcement Learning



To enable machines to learn this visual reasoning behavior, the researchers developed a reinforcement learning framework called Visual Planning via Reinforcement Learning (VPRL).



This framework trains large vision models using GRPO-based optimization so that they improve their planning ability through trial and feedback.

The system was evaluated on several benchmark environments including FrozenLake,

Maze, and MiniBehavior which emphasize navigation and spatial planning.



Results demonstrate that visual planning strategies outperform text-only reasoning approaches in these environments, showing the effectiveness of the visual inference paradigm.

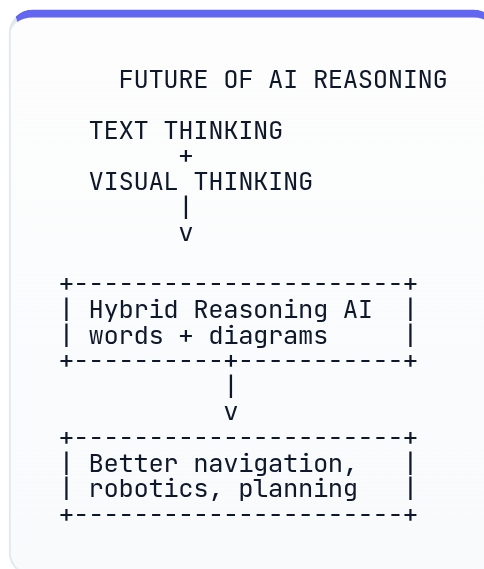
Uses reinforcement learning to optimize visual planning behavior

GRPO enables efficient post-training of large vision models

Benchmarks include FrozenLake, Maze, and MiniBehavior navigation tasks



[horizon] Implications for the Future of AI Reasoning



The success of visual planning suggests that reasoning systems do not need to rely exclusively on language as their thinking medium.



By combining textual and visual reasoning channels, AI models can adapt their thinking style to the structure of the task.